JSGP · SHINE HACKATHON

# Morning Triage

The morning triage for One Care caseworkers:
who needs you first, and why, in plain language.

THE PROBLEM

# Help that must be asked for

Alarm buttons and pendants only help while the fallen person can still press them. An unconscious senior presses nothing.

The fall becomes a **long lie**: hours helpless on the floor before anyone thinks to check.

THE APPROACH

# The home already knows

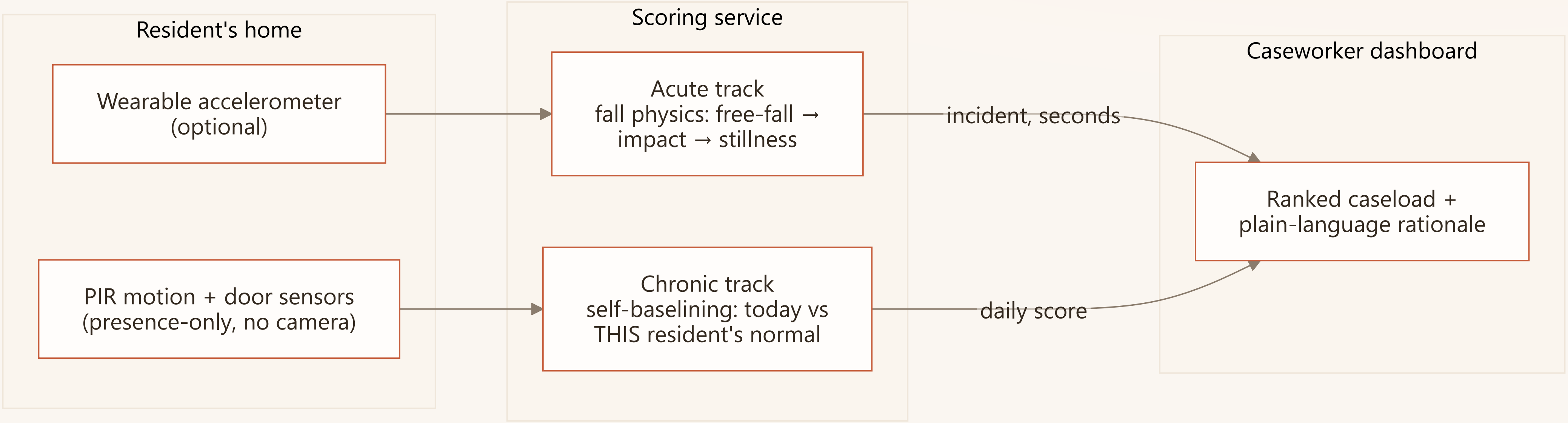
Routine is a vital sign, and absence of motion is data.
The ambient track asks the senior to wear nothing, press nothing, and puts no camera in the room.

One optional bangle adds a second layer:
falls caught in seconds instead of hours.

All of it is off-the-shelf hardware, and Singapore already deploys senior alert wearables at 26,800-resident scale.

HOW IT'S BUILT

# The three **layers**




The home senses. The service scores. The caseworker decides.

HOW IT WORKS

# A fall has a **signature**

WHAT THE SENSOR SAW


Replayed real recording · the detector requires this exact order — free-fall, impact, stillness

Straight from the product: the drill-down shows the signal behind every incident

The detector requires this **ordered sequence**: a free-fall dip, an impact spike within half a second, then stillness. A dropped phone cannot fake the order.

THE TWO DATASETS

# Why these two **recordings**

| 4,505 | 220 days |
| --- | --- |
| REAL RECORDINGS · 1,798 FALLS · 38 PEOPLE · SISFALL | ONE SENIOR, LIVING ALONE · CASAS ARUBA |

One teaches the detector what a fall is in physics.
The other teaches the system what one person's normal life is in rhythm.

Both run on the same cheap sensor class a worn bangle carries, the **ADXL345** accelerometer, so what the detector sees in the data is what it would see in a home.

WHERE THE NUMBER COMES FROM

# How we got our number

- We wrote the detector first, the dip, spike, stillness rule.
- Then we ran it over all **4,505** recordings and **counted**: falls caught, and ordinary activities that false-fired.
- Then we turned the threshold dials until it caught **96.2%** of the falls.

The number comes from counting an experiment, not from a claim. We measured a detector and did not train a model, so it carries to Singapore with no local data.

WHAT THE NUMBERS SAY

# We ran it on real recordings

**96.2%**
OF 1,798 REAL FALLS DETECTED · SISFALL

**35.2% → 18.8%**
FALSE ALARMS: YOUNG ADULTS → ELDERLY MOVEMENT

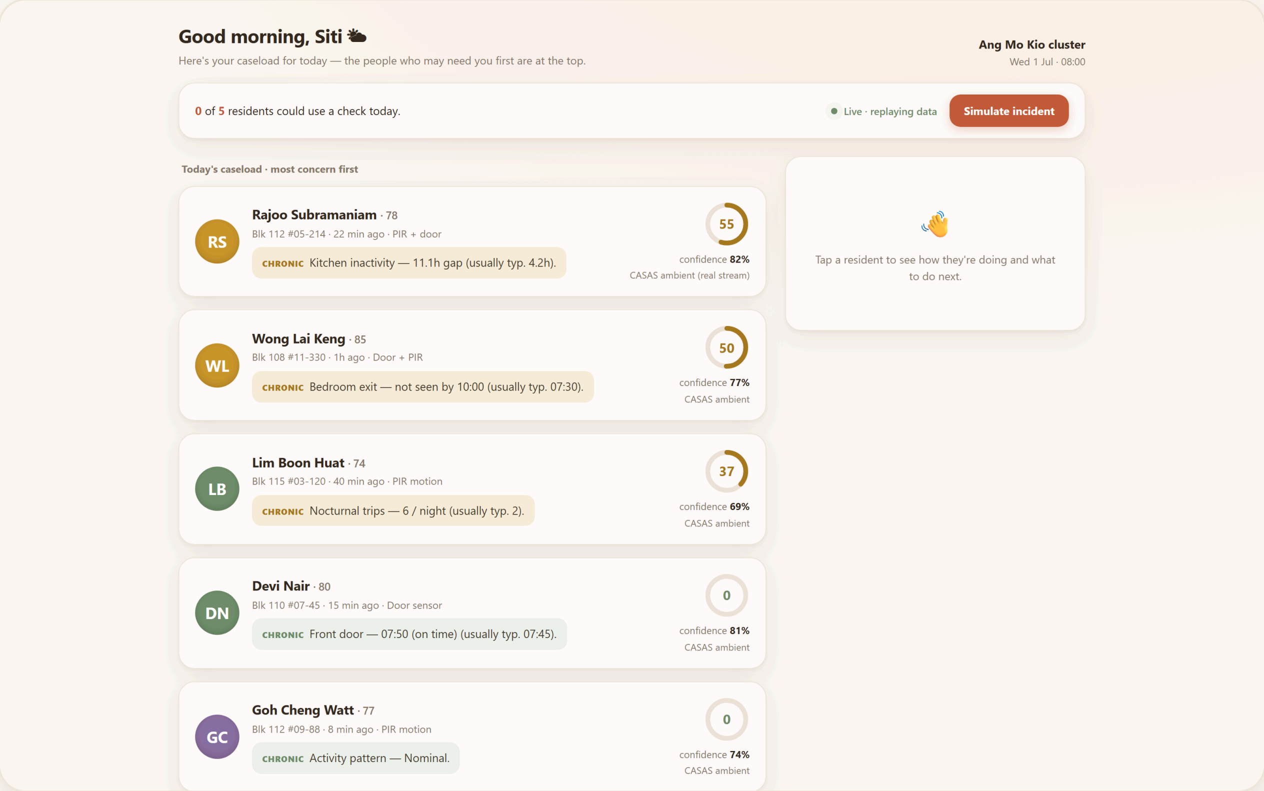
**220 days**
ONE REAL HOME · CASAS ARUBA

The detector caught 96.2% of the 1,798 real falls. The false alarms split by body: 35.2% on young adults hurling themselves into lab falls, but **18.8%** on the elderly participants, roughly half, because older movement is slower and less abrupt. On elderly falls the detector caught **84.0%**, though all 75 come from a single volunteer, so we hold that figure lightly. Nothing is trained here, because the thresholds came straight out of the data.

**Rajoo Subramaniam** · 78

THE PRODUCT

# One screen, calm by default




Every resident is ranked by need, and every row explains itself in one sentence.

EXPLAINABLE BY CONTRACT

# Every number **decomposes**

**WHY SHE'S RANKED HERE**

| | |
|---|---|
| Peak impact<br>walking < 1.4 g | **11.2 g** |
| Post-impact stillness | **12 s** |
| Orientation | **Horizontal** |

Every score breaks into the sensor facts that produced it. Nothing is taken on faith

| | | |
|---|---|---|
| **100**<br>ACUTE RISK | **100%**<br>CONFIDENCE | **Accelerometer**<br>SOURCE |

A stale sensor lowers confidence. It never inflates risk, so a dead sensor cannot fake a crisis

The system ranks and explains.

**A human decides.**

THE DEMO BEAT

# A fall preempts everything

**0** of **5** residents could use a check today.

Live · replaying data

**Simulate incident**

Each press streams a different real recorded fall through the live detector, the same signal a worn bangle would send. A second button injects a dropped phone the detector ignores

**FALL DETECTED**

TA

**Tan Ah Moi** · 82

Blk 108 #04-210 · just now · Accelerometer

ACUTE Fall detected — 11.2 g impact then 12 s no movement.

**100**

confidence **100%**

Wearable bangle

**Call now** **Escalate**

The fall pins to the top in seconds and states its evidence in plain words

### WHAT TO DO NEXT

CALL NOW. If no response, escalate to MOH 24/7 line + dispatch.

The system suggests the next step. The caseworker makes the call

HONEST LIMITS

# What we **don't** claim

- It shrinks time to detection. It does not promise to catch every event.
- The senior false-alarm rate is **measured** rather than guessed, because it came in at 18.8% on real elderly recordings, lower than young adults. What stays thin is elderly fall data, since our falls come from one volunteer, and a lab fall is not the same as a home fall.
- A missed fall (or a bangle left on the nightstand) degrades to the ambient track, where hours of stillness still surface that resident by morning.
- An irregular life earns wide baselines. The system says so through low confidence instead of hiding it.

THE NEXT STEP

# What a pilot would **prove**

- **Escalation that completes the loop.** Automated welfare calls, and a handoff to the MOH/AIC line when a fall goes unacknowledged.
- **A morning brief per flagged resident.** Auto-drafted wording, grounded strictly in the deterministic features. The drafting layer never touches a score.
- **One One Care cluster to start.** Measured on two numbers: time to detection, and caseworker minutes saved per shift.

THE ASK

# Pilot with us

One cluster. Commodity sensors. A working system,
validated on **4,505 real recordings** and a **220-day real home**.

Built at SHINE Hackathon. The demo runs live on real recordings, a different one each press.

BACKUP · FOR Q&A

# Why no machine learning

- **No Singapore training data exists.** A model trained on Colombian or American recordings would import exactly the prior we refuse to assume.
- **A fall is physics.** The free-fall, impact and stillness sequence transfers across bodies and homes without any training step.
- **A caseworker must be able to ask why.** Decomposable rules survive that question. A black box does not.
- Where a model helps later: smoothing the wording of the morning brief. It will never produce a score.

BACKUP · FOR Q&A

# Where the **data goes**

What each dataset teaches is earlier in the deck. This is the provenance behind those numbers.

| DATASET | SOURCE | PROVENANCE |
|---|---|---|
| **SisFall** | Universidad de Antioquia | 4,505 recordings, 1,798 real falls, checksum-pinned |
| **CASAS Aruba** | Washington State University | 220-day single-resident home log, checksum-pinned |

Nothing is trained on either dataset. Both are provenance-pinned with checksums in a committed lock file, and every number reproduces from one script.

# From a fall to the dashboard

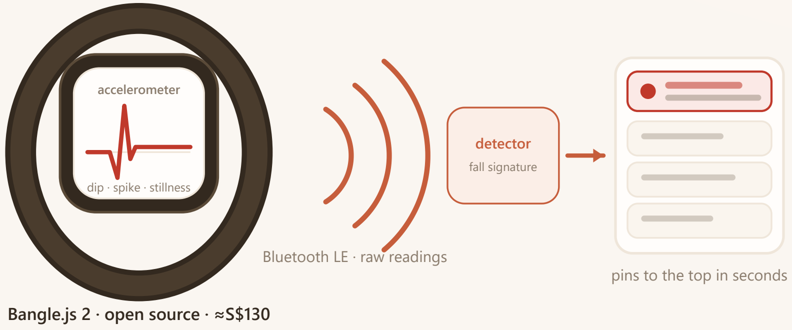
The path is the same in the demo and in deployment. Only the source of the signal changes.

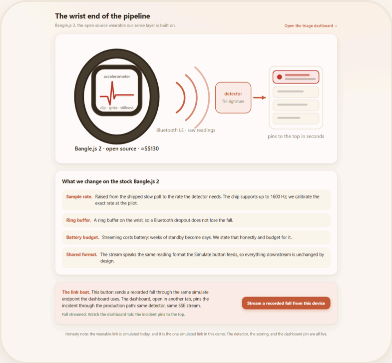
1. The bangle streams accelerometer readings to the scoring service.
2. The detector watches every reading for the ordered fall signature: dip, spike, stillness.
3. A detection pushes an incident event down a live stream the dashboard is always listening to. The resident pins to the top in seconds.

Step 1 is the one link we simulate; there is no wristband in the room.
Each press of the Simulate button feeds a different real SisFall trace straight into step 2; a second button injects a dropped phone the detector ignores. Everything after (detection, push, re-rank) is the production path.

BACKUP · FOR Q&A

# The hardware **exists today**




Live in the dashboard today: /hardware streams a recorded fall through the production path

| DEVICE | REAL PRODUCT, BUYABLE NOW | PRICE |
|---|---|---|
| Wearable: raw accelerometer, BLE | **Bangle.js 2**, Kionix KX022, open firmware | ≈S$130 |
| Room presence · door events | **Aqara P1** motion · contact, 5-yr battery | ≈US$20 · 15 |
| Singapore precedent | **GovTech WAAS** (iWOW, 2025): 170 blocks · 26,800 seniors | govt tender |

BACKUP · FOR Q&A

# What a flat **needs**

- Motion sensors in the rooms that carry routine: kitchen, bedroom, bathroom, living room.
- One contact sensor on the front door.
- One optional bangle for the acute track. The routine track works with nothing worn at all.
- An installer fills in a one-page sensor-to-area map. No cameras, no microphones, no rewiring.